Experiment 6: Bagging, Boosting, and Stacked Ensemble Models

Experiment No.	6
Experiment Title	Bagging, Boosting, and Stacked Ensemble Models
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1 Objective

To implement and compare three ensemble learning strategies – Bagging (Decision Tree base learner), Boosting (AdaBoost and Gradient Boosting), and a heterogeneous Stacked Ensemble (SVM, Naive Bayes, Decision Tree with Logistic Regression as the meta learner) – on the Wisconsin Diagnostic Breast Cancer (WDBC) dataset, using 5-fold cross-validation for hyperparameter tuning and standard classification metrics for final evaluation.

2 Problem Statement

Input: 30 numerical features derived from digitized FNA images of breast masses. **Output:** Binary label – **Malignant (M = 1)** or **Benign (B = 0)**. **Task:** Supervised binary classification, solved using three ensemble strategies:

- **Bagging** – multiple Decision Trees trained on bootstrap samples, combined by majority vote.
- **Boosting** – AdaBoost and Gradient Boosting, trained sequentially to correct prior errors.
- **Stacking** – SVM, Naive Bayes, and Decision Tree as base learners; Logistic Regression as meta learner.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC)
Source	UCI ML Repository / `sklearn.datasets.load_breast_cancer`
Samples	569
Features	30 numerical features
Classes	Benign = 357, Malignant = 212
Train / Test Split	455 / 114 (80% / 20%), `random_state=42`, `stratify=y`
Missing Values	None (0 found)

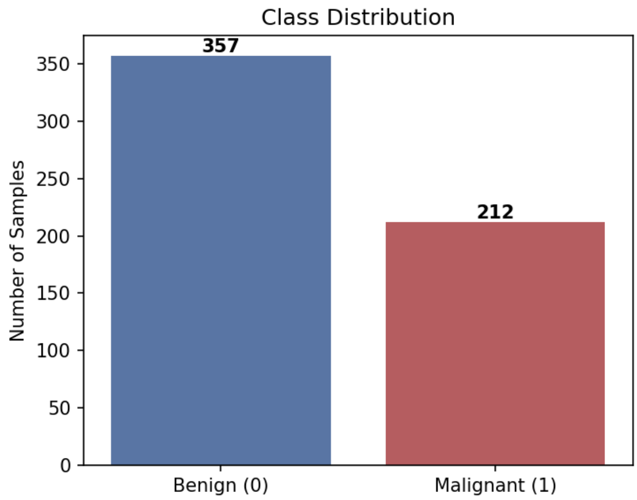

Figure 1: Class distribution: 357 Benign vs. 212 Malignant samples.

The classes are moderately imbalanced (≈63% Benign, 37% Malignant), so stratified sampling was used for the train-test split and for cross-validation folds.

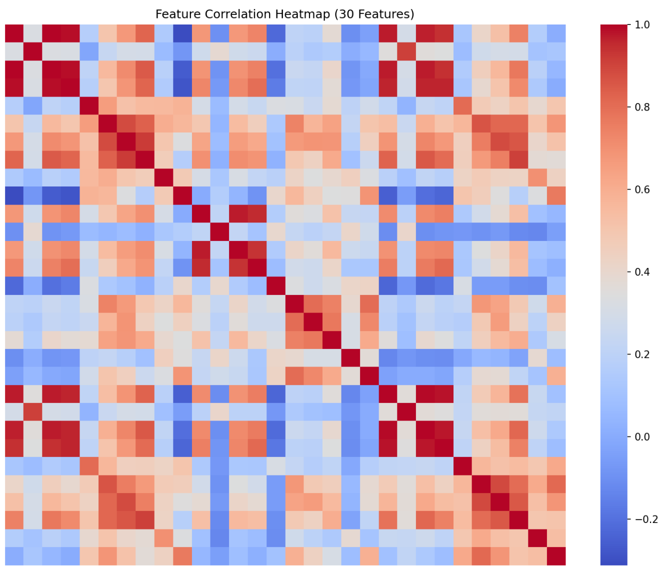

Figure 2: Correlation heatmap of the 30 WDBC features.

Several radius-, perimeter-, and area-based features are strongly correlated, as expected since they describe related geometric properties of the same cell nuclei.

4 Data Preprocessing

- Checked for missing values – none found.
- Target encoded as B $\rightarrow$ 0, M $\rightarrow$ 1.
- 80/20 stratified train-test split (`random_state=42`).

- Features standardized with `StandardScaler` (fit on train only) for the SVM base learner in Stacking; tree-based models used raw features.

5 Machine Learning Algorithms

5.1 Bagging

Bootstrap Aggregation trains multiple Decision Trees independently on bootstrap resamples of the training data; predictions are combined by majority vote. Tuned over `n_estimators` $\in \{10, 50, 100\}$, `max_samples` $\in \{0.5, 0.8, 1.0\}$, `max_features` $\in \{0.5, 0.8, 1.0\}$. Bagging primarily reduces variance.

5.2 Boosting

AdaBoost and Gradient Boosting were both tuned; the better model by CV F1-score was selected for final comparison. Search space: `n_estimators` $\in \{50, 100, 150\}$, `learning_rate` $\in \{0.01, 0.1, 1.0\}$ (Gradient Boosting also over `max_depth` $\in \{1, 3, 5\}$). Boosting primarily reduces bias.

5.3 Stacked Ensemble

```
          Training Data
   +-----------+-----------+
   |           |           |
  SVM      Naive Bayes   Decision Tree
   |           |           |
   +-----------+-----------+
               |
       Logistic Regression (meta learner)
               |
         Final Prediction
```

Base learners were combined via 5-fold internal stacking CV; the meta learner (Logistic Regression) learns how to weight the three base predictions.

6 Hyperparameter Selection (5-Fold CV)

Table 2: Best Configuration per Model (GridSearchCV, scoring = F1)

Model	Best Hyperparameters	CV F1-score
Bagging	n_estimators=50, max_samples=1.0, max_features=0.5	0.958
AdaBoost	n_estimators=150, learning_rate=1.0	0.961
Gradient Boosting	n_estimators=150, learning_rate=1.0, max_depth=1	**0.967**
Stacking (SVM+NB+DT $\rightarrow$ LR)	default base learners, 5-fold internal CV	0.964

Gradient Boosting outperformed AdaBoost on cross-validation F1-score (0.967 vs. 0.961) and was therefore selected as the representative Boosting model for final test-set evaluation.

7 Results

Table 3: Final Test-Set Performance (114 held-out samples)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Bagging	0.965	1.000	0.905	0.950	0.995
Boosting (GB)	0.965	1.000	0.905	0.950	0.988
Stacked Ensemble	**0.974**	1.000	**0.929**	**0.963**	**0.997**

All three models achieved perfect precision on the test set (no benign sample was misclassified as malignant), and the Stacked Ensemble achieved the highest recall, F1-score, accuracy, and AUC.

8 Result Visualization

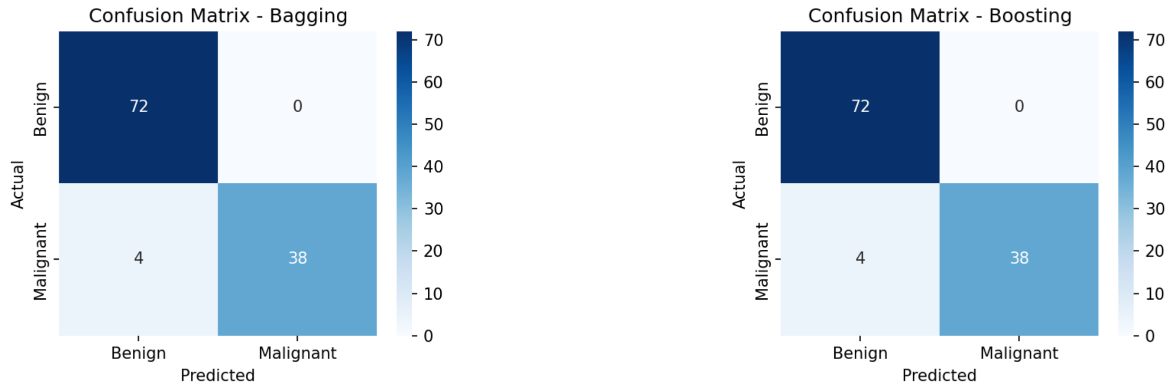

Figure 3: Confusion matrices – Bagging (left) and Boosting (right). Both misclassify 4 malignant cases as benign.

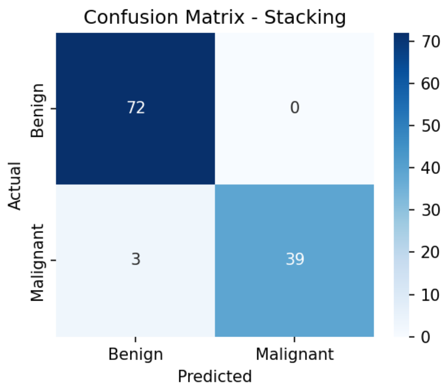

Figure 4: Confusion matrix – Stacked Ensemble. Only 3 malignant cases misclassified as benign, the fewest of all models.

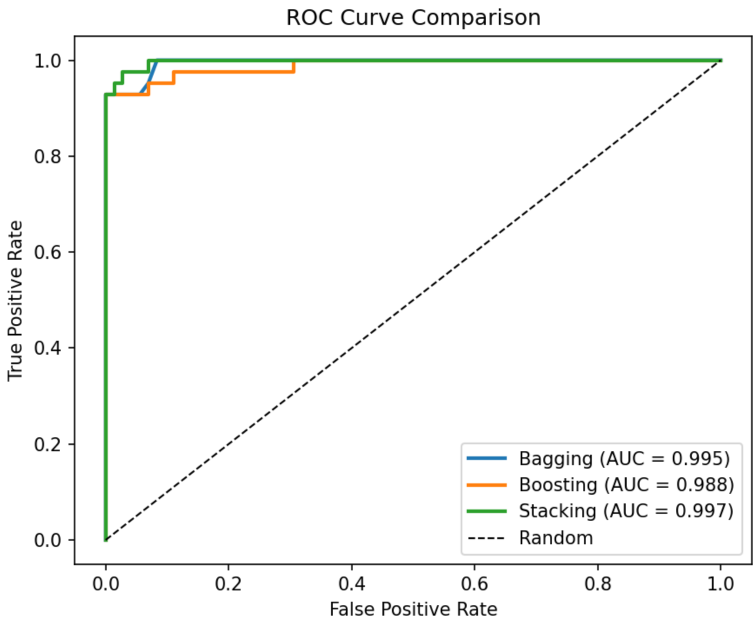

Figure 5: ROC curve comparison. The Stacked Ensemble (AUC = 0.997) dominates Bagging (AUC = 0.995) and Boosting (AUC = 0.988) across nearly all thresholds.

9 Discussion

Bias-variance perspective: Bagging reduces variance by averaging many independently trained Decision Trees over bootstrap samples, stabilizing predictions that would otherwise be sensitive to the specific training subset. Boosting instead reduces bias by sequentially fitting learners to the residual errors of previous rounds – Gradient Boosting's shallow (`max_depth=1`) trees with `learning_rate=1.0` converged to the best CV F1-score among boosting variants. Stacking benefits from combining heterogeneous inductive biases: SVM's margin-based boundary, Naive Bayes' probabilistic independence assumption, and the Decision Tree's axis-aligned splits each capture different structure in the data, and the Logistic Regression meta learner learns to weight these complementary signals.

Generalization: Cross-validation and test performance are close for all three models (e.g., Stacking: CV F1 = 0.964 vs. test F1 = 0.963), indicating good generalization with minimal overfitting.

10 Observation Questions

How does Bagging reduce variance? By training multiple Decision Trees on independent bootstrap samples and aggregating via majority vote, individual trees' overfitting tendencies are averaged out, producing a more stable classifier.

How does Boosting address model bias? Boosting fits learners sequentially, with each new learner focusing on the errors of the ensemble so far; this iterative error-correction reduces systematic bias that a single weak learner would retain.

Why does Stacking benefit from heterogeneous models? SVM, Naive Bayes, and Decision Tree make different structural assumptions about the data, so their errors are only partially correlated. The meta learner exploits this by learning an optimal combination, which is why Stacking achieved the best overall test-set F1-score (0.963) in this experiment.

Which ensemble method performed best and why? The **Stacked Ensemble** performed best (Accuracy = 0.974, F1 = 0.963, AUC = 0.997), because combining heterogeneous base learners captured complementary decision boundaries that neither Bagging (variance reduction alone) nor Boosting (bias reduction alone) could match individually.

11 Conclusion

Bagging, Boosting (AdaBoost / Gradient Boosting), and a Stacked Ensemble (SVM + Naive Bayes + Decision Tree with a Logistic Regression meta learner) were implemented and tuned via 5-fold cross-validation on the WDBC dataset. On the held-out test set, all models exceeded 96% accuracy and 0.98 AUC, with the Stacked Ensemble achieving the best overall performance (Accuracy = 97.4%, F1 = 0.963, AUC = 0.997), confirming that combining heterogeneous base learners can outperform single-family ensembles for this classification task.

12 References

[1] scikit-learn developers, "Ensemble Methods," https://scikit-learn.org/stable/modules/ensemble.html

[2] scikit-learn developers, "BaggingClassifier," https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.BaggingClassifier.html

[3] scikit-learn developers, "AdaBoostClassifier," https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.AdaBoostClassifier.html

[4] scikit-learn developers, "StackingClassifier," https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.StackingClassifier.html

[5] UCI Machine Learning Repository, "Breast Cancer Wisconsin (Diagnostic) Dataset," https://archive.ics.uci.edu